De-rendering, Reasoning, and Repairing Charts with Vision-Language Models

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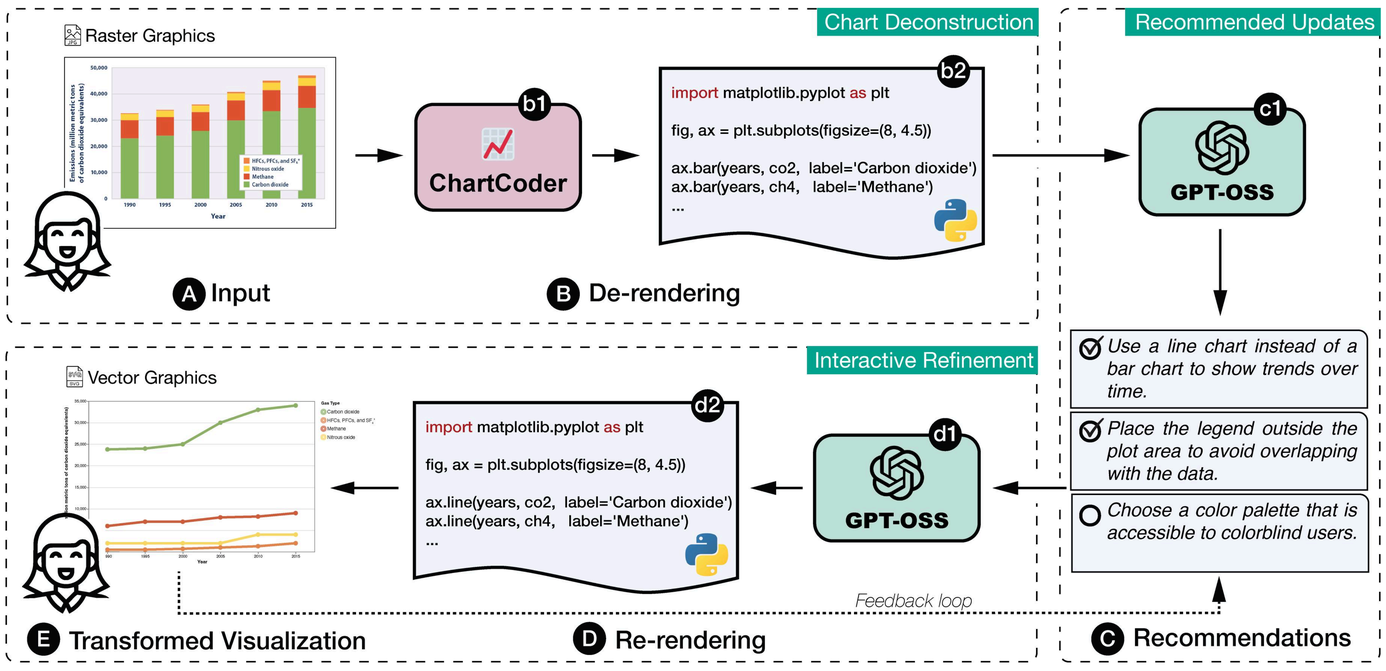
Motivation

Charts play a critical role in science, journalism, and everyday decision-making, yet their effectiveness depends heavily on design quality.

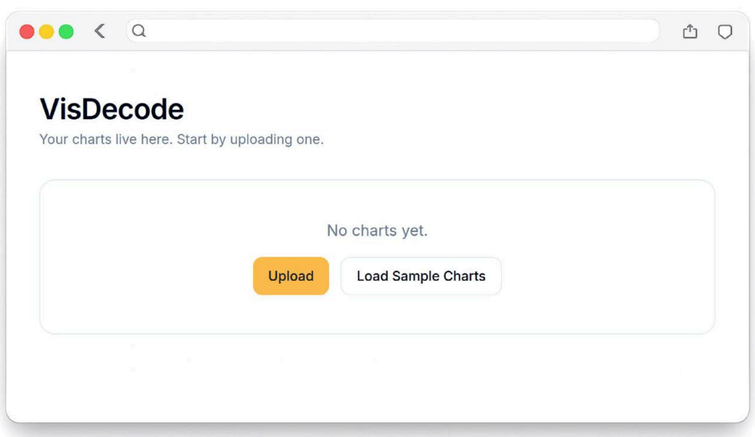
Current tools fall **short**: visualization linters only detect predefined rule violations, and general LLMs/VLMs lack the **visualization literacy** needed to consistently evaluate charts from images.



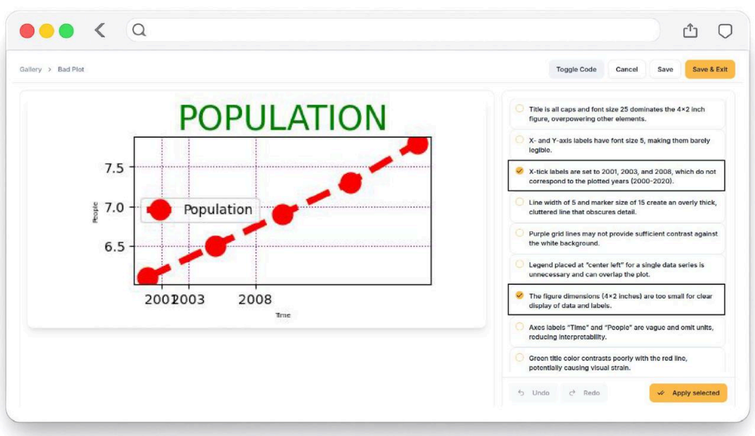
There is a need for a **system** that can interpret a **chart** directly from an image, reason about design principles, and offer actionable, justified improvements.

Framework

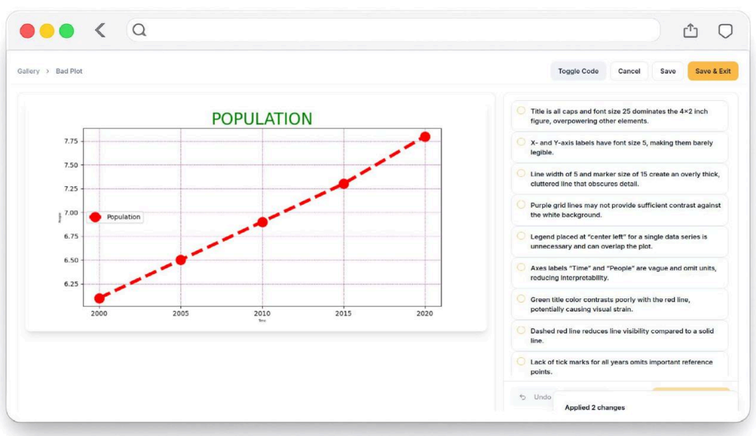
- **Chart Deconstruction**: The system converts a chart image into executable Python/Matplotlib code using **ChartCoder**.
- **Recommended Updates:** Using open-source LLMs, the recovered code is analyzed to identify design issues such as color contrast, axis formatting, readability, or chart-type appropriateness.
- **Interactive Refinement:** Users select which suggestions to apply. The system updates the code, re-renders the chart, and performs a new round of analysis.



A Visualization Upload



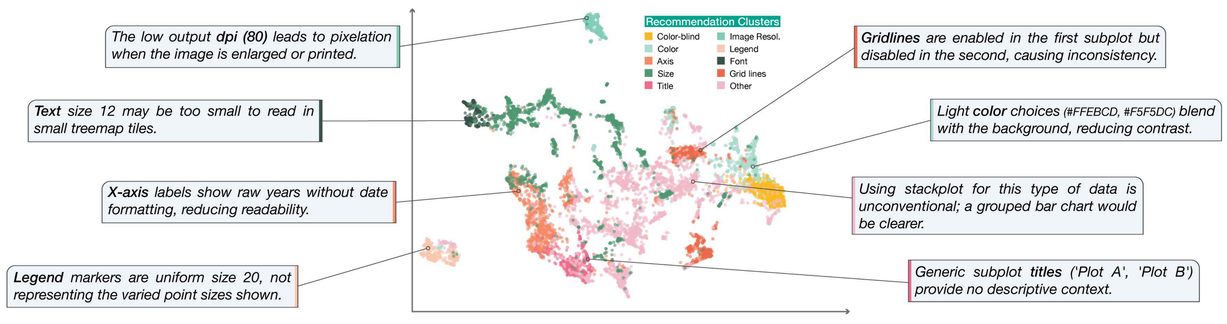
B Recommendation Selection



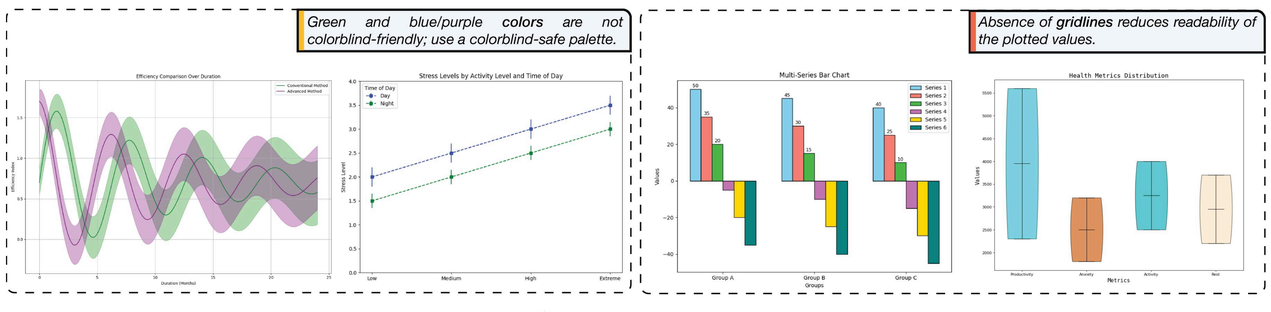
C Updated Visualization

Analysis

- The system was evaluated on **1,000** charts from the **Chart2Code** benchmark. From the de-rendered code, the models generated **10,452** recommendations, demonstrating scalability across diverse chart types.
- The recommendations were embedded in a high dimensional space and then grouped into ten coherent clusters.
- The structure of these clusters shows that the feedback is organized and semantically meaningful. The system identifies relationships between different kinds of design issues and provides consistent and interpretable guidance for improving visualizations.



A UMAP Recommendation Manifold



B Chart Examples